

# GLOBAL WEATHER REPOSITORY

## Advanced Forecasting Report

PM Accelerator — Data Science Technical Assessment  
Advanced Track Submission

|                            |                            |                                       |                                   |
|----------------------------|----------------------------|---------------------------------------|-----------------------------------|
| <b>143,262<br/>Records</b> | <b>41<br/>Raw Features</b> | <b>59<br/>Engineered<br/>Features</b> | <b>2024 – 2026<br/>Date Range</b> |
|----------------------------|----------------------------|---------------------------------------|-----------------------------------|

Target Variable: temperature\_celsius  
Date Range: 2024-05-16 to 2026-05-24

## 1. Executive Summary

This report presents a comprehensive end-to-end weather forecasting pipeline built on the Global Weather Repository dataset. The pipeline covers data cleaning, exploratory data analysis, anomaly detection, feature engineering, climate and spatial analysis, air quality assessment, feature importance evaluation, and multi-model forecasting — spanning classical statistical methods, gradient-boosted machine learning, and deep learning architectures. The best-performing model, LightGBM, achieves an  $R^2$  of 0.961 on the held-out test set, predicting temperature to within 1.43°C mean absolute error.

|                 |                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------|
| Dataset         | 143,262 rows × 41 columns; no columns dropped; zero exceed the 50% missing threshold            |
| Date Range      | 2024-05-16 to 2026-05-24 — two full years of global readings                                    |
| Target Variable | temperature_celsius — daily readings across cities worldwide                                    |
| Anomalies       | 610 consensus anomalies detected (Isolation Forest + LOF agreement)                             |
| Top Predictors  | temp_roll7, temp_lag1, day_of_year — account for >60% of predictive power                       |
| Best Model      | LightGBM: MAE = 1.431°C, RMSE = 1.955°C, $R^2$ = 0.961                                          |
| Deep Learning   | LSTM $R^2$ = 0.937; Transformer $R^2$ = 0.931                                                   |
| Air Quality     | PM2.5 and CO rise with temperature and fall with wind speed — hot stagnant air traps pollutants |

## 2. Data Cleaning & Preprocessing

The raw dataset arrived in excellent condition — all 41 columns were retained because none exceeded the 50% missing-value threshold. The last\_updated column was parsed as datetime and the dataframe sorted chronologically to preserve temporal ordering throughout the pipeline.

|                  |                                                                              |
|------------------|------------------------------------------------------------------------------|
| Missing Values   | Numeric nulls filled with per-country median; global median used as fallback |
| Outlier Capping  | Core features IQR-capped at 1st–99th percentile to suppress extremes         |
| Datetime Parsing | last_updated parsed and dataframe sorted chronologically                     |
| Columns Dropped  | None — all 41 original columns retained                                      |
| Final Shape      | 143,262 rows × 41 columns (pre feature engineering)                          |

## 3. Exploratory Data Analysis

Exploratory analysis examined feature distributions, pairwise correlations, and the global daily average temperature trend across the two-year observation window. Seasonal decomposition with a weekly period revealed a non-stationary global series requiring differencing for ARIMA.

### Key Correlations

|                                   |                                                        |
|-----------------------------------|--------------------------------------------------------|
| temperature_celsius ↔ feelslike_c | Near-perfect positive correlation ( $r \approx 0.99$ ) |
|-----------------------------------|--------------------------------------------------------|

|                        |                               |
|------------------------|-------------------------------|
| pressure_mb ↔ wind_kph | Weak negative correlation     |
| humidity ↔ precip_mm   | Moderate positive correlation |

Stationarity Test

The Augmented Dickey-Fuller (ADF) test on the global daily average temperature series returned a statistic of  $-1.237$  and a p-value of  $0.658$ , confirming **non-stationarity**. First-order differencing ( $d = 1$ ) was applied prior to fitting ARIMA.

4. Anomaly Detection

Two independent anomaly detectors were applied; a data point was flagged only when **both** detectors agreed (consensus approach). This conservative strategy minimises false positives while surfacing genuine extremes.

| Detector                    | Contamination | Points Flagged |
|-----------------------------|---------------|----------------|
| Isolation Forest            | 3%            | 4,298          |
| Local Outlier Factor (k=20) | 3%            | 4,298          |
| Consensus (both agree)      | —             | 610            |

The 610 consensus anomalies are concentrated around extreme seasonal spikes and unusual inter-day temperature jumps — consistent with physically plausible but rare meteorological events.

5. Feature Engineering

18 new features were derived from the original 41, expanding the feature space to 59 columns. Features span temporal signals, rolling statistics, lag values, and physically-motivated derived indices designed to capture both short-term autocorrelation and seasonal patterns.

| Category         | Features Created                                      |
|------------------|-------------------------------------------------------|
| Temporal         | year, month, day_of_year, week, season                |
| Rolling Averages | temp_roll7, temp_roll30, humidity_roll7, precip_roll7 |
| Lag Features     | temp_lag1, temp_lag3, temp_lag7                       |
| Derived Indices  | heat_index, feels_delta                               |

Final dataset shape after engineering: **143,262 × 59 features**.

6. Climate & Spatial Analysis

Country-level aggregation revealed wide variation in mean temperatures across the globe. Hottest nations are predominantly in equatorial and desert regions; coldest nations lie at high latitudes. The top-15 country comparison confirms the dataset spans a broad climate spectrum — essential for a generalised forecasting model. This diversity strengthens the model's ability to generalise across climate zones without requiring region-specific retraining.

## 7. Environmental Impact — Air Quality

Eight air quality metrics are present in the dataset. Cross-correlation analysis between air quality indicators and meteorological variables surfaces an important environmental relationship between stagnant heat and pollutant concentration.

|                  |                                                                                                                                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Columns Analysed | air_quality_Carbon_Monoxide, air_quality_Ozone, air_quality_Nitrogen_dioxide, air_quality_Sulphur_dioxide, air_quality_PM2.5, air_quality_PM10, air_quality_us-epa-index, air_quality_gb-defra-index |
| Key Finding      | PM2.5 and Carbon Monoxide show positive correlation with temperature and negative correlation with wind speed                                                                                        |
| Implication      | Hot, stagnant air traps pollutants near the surface — directly relevant for urban air quality forecasting and public health modelling                                                                |

## 8. Feature Importance

Feature importance was assessed via two complementary methods: Random Forest Mean Decrease in Impurity (MDI) and SHAP values derived from the XGBoost model. Both methods converge on the same top three predictors, providing strong cross-method validation.

| Rank | Feature     | Description                       | Method Consensus |
|------|-------------|-----------------------------------|------------------|
| 1    | temp_roll7  | 7-day rolling average temperature | RF MDI + SHAP    |
| 2    | temp_lag1   | Previous-day temperature          | RF MDI + SHAP    |
| 3    | day_of_year | Cyclical seasonal signal          | RF MDI + SHAP    |

These three features account for more than 60% of predictive power. The dominance of rolling averages and lag features underscores the strong temporal autocorrelation in temperature data. The SHAP beeswarm plot confirmed monotonic positive relationships for all three top features.

## 9. Classical Forecasting

Two classical time-series models were fitted to the 30-day global average temperature series and serve as interpretability baselines. Both are evaluated on a univariate held-out window which differs from the full ML test set — direct metric comparison is therefore not valid.

### ARIMA (2, 1, 2)

Slow ACF decay confirmed non-stationarity, leading to  $d = 1$  differencing. AR and MA orders of 2 were selected via PACF inspection.

MAE: 1.084 | RMSE: 1.240 |  $R^2$ : -2.931

### Prophet — Yearly + Weekly Seasonality

Facebook Prophet was configured with both yearly and weekly seasonal components, achieving the lowest MAE of all models on the 30-day window. Its negative  $R^2$  signals it is outperformed by ML models on the full multivariate test set.

MAE: 0.640 | RMSE: 0.746 | R<sup>2</sup>: -0.423

## 10. Machine Learning Forecasting

Four gradient-boosted and ensemble ML models were trained on the full 59-feature dataset using a strict chronological train/test split to prevent any form of data leakage.

Train: 113,328 samples · Test: 28,333 samples (chronological, no shuffling)

| Model             | MAE (°C) | RMSE (°C) | R <sup>2</sup> |
|-------------------|----------|-----------|----------------|
| LightGBM          | 1.431    | 1.955     | 0.961          |
| XGBoost           | 1.453    | 1.982     | 0.960          |
| Gradient Boosting | 1.489    | 2.032     | 0.958          |
| Random Forest     | 1.490    | 2.063     | 0.957          |

LightGBM achieves the lowest MAE (1.431°C) and RMSE (1.955°C) while recording the highest R<sup>2</sup> (0.961). All four models perform within a narrow 0.06°C MAE band, demonstrating that the feature engineering is robust and consistent across different algorithms.

## 11. Deep Learning Forecasting

Two deep learning architectures were trained on 30-day sliding-window sequences to predict a 7-day horizon. Training comprised 113,300 sequences × 30 timesteps × 14 features.

### LSTM Architecture

LSTM(128) → Dropout(0.2) → LSTM(64) → Dropout(0.2) → Dense(32) → Dense(7)

### Temporal Transformer Architecture

Input → Dense(64) → Positional Encoding → MultiHeadAttention(4 heads) → LayerNorm → FFN(128) → LayerNorm → GlobalAvgPool → Dense(64) → Dense(7)

| Model       | MAE (°C) | RMSE (°C) | R <sup>2</sup> |
|-------------|----------|-----------|----------------|
| LSTM        | 1.830    | 2.484     | 0.937          |
| Transformer | 1.923    | 2.610     | 0.931          |

Both models exceed R<sup>2</sup> = 0.93, confirming competitive performance. They trail the best ML model by approximately 0.4°C MAE. The Transformer is slightly weaker but offers enhanced interpretability through attention weights.

## 12. Ensemble Model & Final Leaderboard

A weighted ensemble was constructed from the four ML models using inverse-MAE weighting — models with lower error receive proportionally higher influence on the final prediction.

### Ensemble Weights

| Model             | Inverse-MAE Weight |
|-------------------|--------------------|
| LightGBM          | 0.2560             |
| XGBoost           | 0.2522             |
| Gradient Boosting | 0.2460             |
| Random Forest     | 0.2458             |

Ensemble Result — MAE: 1.446 | RMSE: 1.982 | R<sup>2</sup>: 0.960

### Final Model Leaderboard

| Rank | Model               | MAE (°C) | RMSE (°C) | R <sup>2</sup> |
|------|---------------------|----------|-----------|----------------|
| 1    | LightGBM            | 1.431    | 1.955     | 0.961          |
| 2    | Ensemble (Weighted) | 1.446    | 1.982     | 0.960          |
| 3    | XGBoost             | 1.453    | 1.982     | 0.960          |
| 4    | Gradient Boosting   | 1.489    | 2.032     | 0.958          |
| 5    | Random Forest       | 1.490    | 2.063     | 0.957          |
| 6    | LSTM                | 1.830    | 2.484     | 0.937          |
| 7    | Transformer         | 1.923    | 2.610     | 0.931          |
| 8 *  | Prophet             | 0.640    | 0.746     | −0.423         |
| 9 *  | ARIMA               | 1.084    | 1.240     | −2.931         |

\* Classical models evaluated on the 30-day univariate series only (different test set).

## 13. Key Insights

|                   |                                                                        |
|-------------------|------------------------------------------------------------------------|
| Dataset Quality   | 143,262 records, zero columns dropped — exceptionally clean            |
| Temporal Coverage | Two full years (May 2024 – May 2026) across global cities              |
| Anomalies         | 610 consensus anomalies concentrated in extreme seasonal spikes        |
| Top Features      | Rolling-7 temp, lag-1 temp, day_of_year drive >60% of predictive power |
| Air Quality       | PM2.5 and CO rise with temperature, fall with wind speed               |
| Best Single Model | LightGBM: MAE = 1.431°C, RMSE = 1.955°C, R <sup>2</sup> = 0.961        |

|                       |                                                                             |
|-----------------------|-----------------------------------------------------------------------------|
| <b>Ensemble Value</b> | Marginal gain over LightGBM alone; adds stability, not accuracy             |
| <b>Deep Learning</b>  | Competitive at $R^2 \geq 0.931$ but lags ML by $\sim 0.4^\circ\text{C}$ MAE |
| <b>Stationarity</b>   | Global avg temp non-stationary (ADF $p = 0.658$ ); differencing required    |
| <b>Scalability</b>    | Full pipeline processes 143K rows on CPU — production-ready                 |

## 14. Future Work

|                                  |                                                                   |
|----------------------------------|-------------------------------------------------------------------|
| <b>Hyperparameter Tuning</b>     | Optuna / Bayesian search for LightGBM and Transformer             |
| <b>Multi-Target Forecasting</b>  | Jointly predict temperature, humidity, and precipitation          |
| <b>Probabilistic Forecasting</b> | Quantile regression or MC-Dropout for uncertainty bands           |
| <b>Real-Time Pipeline</b>        | Stream live data via OpenWeatherMap API and retrain incrementally |
| <b>Deployment</b>                | Wrap best model as FastAPI endpoint with Streamlit dashboard      |
| <b>Regional Models</b>           | Train separate models per climate zone for localised accuracy     |

## 15. Technology Stack

|                         |                                                                        |
|-------------------------|------------------------------------------------------------------------|
| <b>Data Processing</b>  | Pandas, NumPy, SciPy                                                   |
| <b>Visualisation</b>    | Matplotlib, Seaborn, Plotly                                            |
| <b>Machine Learning</b> | Scikit-learn, XGBoost, LightGBM, SHAP                                  |
| <b>Time-Series</b>      | Statsmodels, Prophet                                                   |
| <b>Deep Learning</b>    | TensorFlow / Keras                                                     |
| <b>Notebook</b>         | Jupyter, nbformat                                                      |
| <b>Dataset</b>          | Global Weather Repository — Kaggle (143,262 rows $\times$ 41 features) |

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**Submitted for PM Accelerator — Data Scientist / Analyst Role**

Advanced Track · Global Weather Repository Assessment

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